

8.6 Consider the wage equation

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + e_i \quad (XR8.6a)$$

where wage is measured in dollars per hour, education and experience are in years, and $METRO = 1$ if the person lives in a metropolitan area. We have $N = 1000$ observations from 2013.

- a. We are curious whether holding education, experience, and $METRO$ constant, there is the same amount of random variation in wages for males and females. Suppose $\text{var}(e_i | \mathbf{x}_i, FEMALE = 0) = \sigma_M^2$ and $\text{var}(e_i | \mathbf{x}_i, FEMALE = 1) = \sigma_F^2$. We specifically wish to test the null hypothesis $\sigma_M^2 = \sigma_F^2$ against $\sigma_M^2 \neq \sigma_F^2$. Using 577 observations on males, we obtain the sum of squared OLS residuals, $SSE_M = 97161.9174$. The regression using data on females yields $\hat{\sigma}_F = 12.024$. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.

$$N_M = 577, k = 4 \Rightarrow df_M = 573$$

$$N_F = 423, k = 4 \Rightarrow df_F = 419$$

$$SSE_M = 97161.9174, \hat{\sigma}_F = 12.024$$

$$\hat{\sigma}_M^2 = \frac{SSE_M}{df_M} \approx 169.567$$

$$\hat{\sigma}_F^2 = (12.024)^2 \approx 144.577$$

$$H_0: \sigma_M^2 = \sigma_F^2$$

$$H_A: \sigma_M^2 \neq \sigma_F^2$$

$$\alpha = 0.05$$

$$\text{test statistics: } F = \frac{\hat{\sigma}_M^2}{\hat{\sigma}_F^2} \stackrel{H_0}{\sim} F(573, 419)$$

$$RR = \left\{ F \leq F_{1-\frac{\alpha}{2}}(n_1-1, n_2-1) = \frac{1}{F_{\frac{\alpha}{2}}(n_2-1, n_1-1)} \text{ or } F \geq F_{\frac{\alpha}{2}}(n_1-1, n_2-1) \right\}$$

$$RR = \left\{ F \leq F_{0.025}(573, 419) = 0.838 \text{ or } F \geq F_{0.975}(573, 419) = 1.197 \right\}$$

$$F_o = \frac{\hat{\sigma}_M^2}{\hat{\sigma}_F^2} \approx 1.1729$$

$$F_o \notin RR, \text{ do not reject } H_0$$

There is no sufficient evidence to conclude that the variance of wages is different between males and females at the 5% significance level, holding $EDUC$, $EXPER$, and $METRO$ constant.

- b. We hypothesize that married individuals, relying on spousal support, can seek wider employment types and hence holding all else equal should have more variable wages. Suppose $\text{var}(e_i | x_i, \text{MARRIED} = 0) = \sigma_{\text{SINGLE}}^2$ and $\text{var}(e_i | x_i, \text{MARRIED} = 1) = \sigma_{\text{MARRIED}}^2$. Specify the null hypothesis $\sigma_{\text{SINGLE}}^2 = \sigma_{\text{MARRIED}}^2$ versus the alternative hypothesis $\sigma_{\text{MARRIED}}^2 > \sigma_{\text{SINGLE}}^2$. We add *FEMALE* to the wage equation as an explanatory variable, so that

$$\text{WAGE}_i = \beta_1 + \beta_2 \text{EDUC}_i + \beta_3 \text{EXPER}_i + \beta_4 \text{METRO}_i + \beta_5 \text{FEMALE} + e_i \quad (\text{XR8.6b})$$

Using $N = 400$ observations on single individuals, OLS estimation of (XR8.6b) yields a sum of squared residuals is 56231.0382. For the 600 married individuals, the sum of squared errors is 100,703.0471. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.

$$N_{\text{SINGLE}} = 400, k = 5 \Rightarrow df_{\text{SINGLE}} = 395$$

$$N_{\text{MARRIED}} = 600, k = 5 \Rightarrow df_{\text{MARRIED}} = 595$$

$$\text{SSE}_{\text{SINGLE}} = 56231.0382, \text{SSE}_{\text{MARRIED}} = 100703.0471$$

$$\hat{\sigma}_{\text{SINGLE}}^2 = \frac{\text{SSE}_{\text{SINGLE}}}{df_{\text{SINGLE}}} \approx 142.357$$

$$\hat{\sigma}_{\text{MARRIED}}^2 = \frac{\text{SSE}_{\text{MARRIED}}}{df_{\text{MARRIED}}} \approx 169.249$$

$$\textcircled{1} H_0: \sigma_{\text{SINGLE}}^2 = \sigma_{\text{MARRIED}}^2$$

$$\textcircled{2} H_a: \sigma_{\text{MARRIED}}^2 > \sigma_{\text{SINGLE}}^2$$

$$\textcircled{3} \alpha = 0.05$$

$$\textcircled{4} \text{test statistic: } F = \frac{\hat{\sigma}_{\text{MARRIED}}^2}{\hat{\sigma}_{\text{SINGLE}}^2} \stackrel{H_0}{\sim} F(595, 395)$$

$$\textcircled{5} \text{RR} = \{F \geq F_{0.95}(595, 395) = 1.165\}$$

$$\textcircled{6} F_0 = \frac{\hat{\sigma}_{\text{MARRIED}}^2}{\hat{\sigma}_{\text{SINGLE}}^2} \approx 1.1889$$

$$\textcircled{7} F_0 \in \text{RR}, \text{ reject } H_0$$

There is sufficient evidence to conclude that the variance of wages for married individuals is greater than that for single individuals at 5% significance level.

- c. Following the regression in part (b), we carry out the NR^2 test using the right-hand-side variables in (XR8.6b) as candidates related to the heteroskedasticity. The value of this statistic is 59.03. What do we conclude about heteroskedasticity, at the 5% level? Does this provide evidence about the issue discussed in part (b), whether the error variation is different for married and unmarried individuals? Explain.

$$N = 1000, \quad g = 4$$

$$\textcircled{1} \quad H_0: \text{homoskedasticity}$$

$$\textcircled{2} \quad H_A: \text{heteroskedasticity}$$

$$\textcircled{3} \quad \alpha = 0.05$$

$$\textcircled{4} \quad \text{test statistic: } NR^2 \stackrel{H_0}{\sim} \chi^2(4)$$

$$\textcircled{5} \quad RR = \{ NR^2 \geq \chi^2_{0.95}(4) = 9.4873 \}$$

$$\textcircled{6} \quad NR^2 = 59.03$$

$$\textcircled{7} \quad NR^2 \in RR, \text{ reject } H_0$$

There is sufficient evidence to conclude that there exists heteroskedasticity at 5% significance level.

However, this test does not specifically show that variance is different between married and single individuals, since it tests all regressors jointly.

- d. Following the regression in part (b) we carry out the White test for heteroskedasticity. The value of the test statistic is 78.82. What are the degrees of freedom of the test statistic? What is the 5% critical value for the test? What do you conclude?

White test:

1. original variables: 4 (EDUC, EXPER, METRO, FEMALE)
2. squared terms: 4 (dummy square is also itself, but it also counts)
3. interaction terms: 6 ($\binom{4}{2}$)

$$\Rightarrow g = 14 \text{ (df)}$$

① H_0 : homoskedasticity

H_A : heteroskedasticity

② $\alpha = 0.05$

③ test statistics: $NR^2 \stackrel{H_0}{\sim} \chi^2(14)$

④ $RR = \{NR^2 \geq \chi^2_{0.95}(14) = 23.68457\}$ (critical value)

⑤ $NR^2 = 78.82$

⑥ $NR^2 \in RR$, reject H_0

There is sufficient evidence to conclude that there exists heteroskedasticity at 5% significance level.

- e. The OLS fitted model from part (b), with usual and robust standard errors, is

$$\widehat{WAGE} = -17.77 + 2.50EDUC + 0.23EXPER + 3.23METRO - 4.20FEMALE$$

(se)	(2.36)	(0.14)	(0.031)	(1.05)	(0.81)
(robse)	(2.50)	(0.16)	(0.029)	(0.84)	(0.80)

For which coefficients have interval estimates gotten narrower? For which coefficients have interval estimates gotten wider? Is there an inconsistency in the results?

(1) Narrower: EXPER, METRO, FEMALE

(2) Wider: Intercept, EDUC

(3) No real inconsistency

The differences arise because robust standard errors adjust for heteroskedasticity, and the direction of change depends on how the error variance is related to each regressor.

f. If we add *MARRIED* to the model in part (b), we find that its t -value using a White heteroskedasticity robust standard error is about 1.0. Does this conflict with, or is it compatible with, the result in (b) concerning heteroskedasticity? Explain.

No, this does not conflict with the result in (b).

The test in part (b) concerns whether the error variances differ between married and single individuals, while the t -test for the coefficient on *MARRIED* examines whether *MARRIED* affects the conditional mean of wages.

It is entirely possible for the error variance to differ across groups (heteroskedasticity) even if *MARRIED* has no significant effect on the mean outcome.